

Code-Layout, Readability and Reusability

Date	03 October 2026
Team ID	SWTID-2026-6945
Project Name	ComicCraft - AI Comic Story Creator using Gemini Models
Maximum Marks	5 Marks

This document evaluates the structure, readability and reusability of the ComicCraft codebase.

Code Layout Checklist

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing used throughout	Yes	4-space PEP 8 indentation in all Python files; 2-space in HTML templates.
2	Proper File Structure	Files and folders logically organised	Yes	app/ (routes, config, schemas), app/services/ (one file per AI or export step), templates/, static/, tests/, scripts/.
3	Meaningful Variable Names	Variables reflect their purpose	Yes	e.g. story_prompt, character_name, image_paths, pdf_path, layout.
4	Function / Method Names	Functions descriptively named	Yes	generate_outline, generate_story, generate_image, build_comic_layout, save_pdf, create_comic (names match the project brief).
5	Code Comments	Inline and block comments explain logic	Yes	Module and function docstrings plus short inline comments for non-obvious logic (retry, JSON cleaning, locks).
6	Modular Design	Code split into reusable functions / modules	Yes	Each workflow step is its own module; pipeline.create_comic() composes them.
7	No Redundant Code	Duplicate or unused code removed	Yes	Form route and JSON route share create_comic() and PromptRequest; one shared Gemini client.
8	Error Handling	Exceptions handled gracefully	Yes	GenerationError, Gemini retries, validation errors (422), 502/500 pages, placeholder image fallback.

Reusable Components / Modules

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	services/gemini_client.call_gemini()	Python, google-genai	gemini_flash and gemini_pro (any other Gemini task)	High
2	services/pipeline.create_comic()	Python	/generate form route and /generate-comic/json API route	High
3	schemas.PromptRequest	Python, Pydantic	Form validation and JSON API validation; auto-generated /docs	High
4	services/image_generator.generate_image()	Python, Diffusers, Pillow	Pipeline (each panel) and /test-image route	High
5	services/exporters.save_pdf()	Python, FPDF2	Comic export; can export any layout list	Medium

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
6	templates (Jinja2) with shared style	HTML, CSS, Jinja2	index, preview and success pages share visual style	Medium

Overall Code Quality Assessment

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Clear separation of routing, services, templates and tests.
Readability	4	Small functions with docstrings; some prompt strings are long.
Reusability	4	Shared client, pipeline and schema; templates duplicate some CSS.
Documentation / Comments	4	README, docstrings and .env.example; no auto-generated API reference beyond /docs.
Overall Score	4.25	Average of the four ratings above (self-assessment).